

AI TALENT INTELLIGENCE & RECRUITMENT ECOSYSTEM

Complete Technical Mapping Blueprint (Including PPT Processing Libraries & Multimodal AI)

Architecture: Node.js Gateway + Python AI Microservice

AI Stack: Groq Llama 3.1 + Gemini 1.5 Flash Vision + Hybrid ML

PPT Stack: python-pptx + pdf2image + Cloudinary Vault

1. TECHNICAL REQUIREMENTS & COMPLETE SOLUTION MAPPING TABLE

Requirement / Module	Feature & Functional Scope	Tech Stack & Architecture	APIs / LLMs / ML Models / Libraries
IDE & Vibe Coding Core	Rapid UI generation, AI composition, and modular code assembly.	v0.dev, Cursor IDE, Next.js App Router	React 18, Tailwind CSS, Lucide Icons, Shadcn UI
Core Architecture Gateway	Microservices orchestration, DB access, REST endpoints, and SSE streaming.	Node.js Express, Python FastAPI	REST APIs, Axios, Socket.io / SSE Protocol
Storage & Vector Store	Profile storage, secure pitch deck uploads, and RAG vector search.	Supabase / Postgres, Cloudinary Vault	Qdrant / FAISS, Prisma ORM, Cloudinary Secure SDK
AI Talent Profile Engine	Auto profile generation from GitHub, LinkedIn, Resumes & Hackathons.	Python Microservice, PyGithub, OAuth 2.0	Groq Llama 3.1 70B, LangChain / LangGraph
AI Talent Score™ System	Multidimensional radar score (Coding, Project Quality, Leadership, Innovation).	Python, NumPy, Pandas Engine	Custom Algorithmic Scoring, Groq LLM Evaluator
Candidate Analytics Dashboard	Talent analytics, skill badges, career insights, and live score updates.	Next.js App Router, Recharts / Canvas	Client-side SSR/CSR, SSE Stream Listener
Candidate Resume Workflow (Upload & Builder)	Upload original candidate resume, extract data, verify claims vs GitHub, and generate ATS Resume & Portfolio Website.	Next.js, Python Parsing Engine, PDFKit / Puppeteer	pypdf / pdfplumber, Groq Llama 3.1 (ATS Rules & Verified Badges)
Career Guidance & Salary Predictor	Skill gap analysis, career roadmaps, and instant ML salary predictions.	Python (Scikit-Learn)	RandomForestRegressor (ML Model), Groq RAG
Recruiter Dashboard & Pipeline	Candidate discovery, shortlisting, recruitment pipeline & analytics.	Next.js, Tailwind, Shadcn UI	Express REST Endpoints, PostgreSQL Aggregations
AI Job Matching Engine	Match %, skill similarity, cultural fit, and project relevance scores.	Python Microservice, Vector DB	Sentence-Transformers (all-MiniLM-L6-v2), Cosine Sim
Recruiter AI Copilot	Natural language search ("Find GenAI developers with hackathon wins").	Python, RAG Architecture	LangChain / LangGraph Agent, Groq Llama 3.1
AI Skill Verification & AST Engine	GitHub repository AST analysis, commit velocity, and team contributions.	PyGithub, Python AST Parser	IsolationForest / XGBoost (Anomaly Detector), Groq
AI Interview Agent (Voice & Video)	Real-time voice interview, behavioral & communication reports, confidence index.	Browser MediaDevices, Python	Groq Whisper API (STT), Face-api.js (Confidence), Web Speech API
AI PPT Analyzer & Presentation Intelligence (Mandatory)	Extract PPT slides/text, parse architecture diagrams, detect AI pitch decks, and calculate Innovation, Feasibility, Business & Pitch Scores.	python-pptx, pdf2image + Poppler, Cloudinary Media Vault	Google Gemini 1.5 Flash Vision API, TF-IDF Similarity, Groq Llama 3.1
Hackathon-to-Hiring Pipeline	Winner analytics, team rankings, innovation scoring, and direct hiring access.	Next.js, Express Gateway	PostgreSQL DB, RAG Search Filtering
Trust & Fraud Prevention System	Detect fake certs, plagiarized repos, duplicate profiles, and AI resumes.	Python Security Pipeline	TF-IDF + Cosine Similarity, Image Hash Check, Presidio PII
Multi-Agent AI Framework (Bonus)	Collaborative agents (Resume, Fraud, Interview, Presentation, Recruiter Agents).	Python Microservice	CrewAI / LangGraph Framework, Groq + Gemini Vision
Privacy, Security & System Design	Data masking, zero-trust PII layer, ephemeral storage auto-delete, job queues.	Redis, BullMQ Queue	Zero-Trust PII Masking Engine, Cloudinary Destroy API

2. MANUAL ENVIRONMENT CONFIGURATION (.ENV FILE SETUP)

```
# 1. Server Configuration
PORT=5000
NODE_ENV=development
FRONTEND_URL=http://localhost:3000

# 2. LLM & AI API Keys (Including Vision & Multimodal)
GROQ_API_KEY=gsk_XXXXXXXXXXXXXXXXXXXXXXXXXXXX
GEMINI_API_KEY=AIzaSyXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
# 3. Database Credentials
DATABASE_URL=postgresql://postgres:password@localhost:5432/talent_db
SUPABASE_URL=https://xxxxxxx.supabase.co
SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...

# 4. Storage & Media (Cloudinary for PPT & Pitch Deck Assets)
CLOUDINARY_CLOUD_NAME=your_cloud_name
CLOUDINARY_API_KEY=123456789012345
CLOUDINARY_API_SECRET=abcdefghijklmnopqrstuvwxyz

# 5. Developer & Social Authentication
GITHUB_CLIENT_ID=0v23xxxxxxxxxxxxx
GITHUB_CLIENT_SECRET=xxxxxxxxxxxxxxxxxxxxxxxxxxxxx
GITHUB_PERSONAL_ACCESS_TOKEN=ghp_xxxxxxxxxxxxxxxxxxxxxxx

# 6. Redis Queue Setup (for BullMQ Background Jobs)
REDIS_HOST=127.0.0.1
REDIS_PORT=6379

# 7. Vector Database Setup
QDRANT_URL=https://xxxxxxx.cloud.qdrant.io
QDRANT_API_KEY=xxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

Continuous Agent Adaptability & Dynamic Feature Expansion Agreement

Agent Protocol Note: This updated PDF explicitly includes the full PPT processing stack (python-pptx, pdf2image, Poppler, Cloudinary) alongside Gemini Vision. All AI agents remain fully synced to adaptively integrate any future features, endpoints, or ML models dynamically into this architecture.